

CUSTOMER VALUE PREDICTION REPORT

Author: Olalemi Olaoluwakintan Emmanuel

Project: Customer Value Prediction

Role: Data Scientist

1. INTRODUCTION

1.1 Project Objective

The objective of this project was to develop an end-to-end machine learning pipeline for predicting customers who are likely to become high-value customers based on their historical purchasing behaviour.

The project combined SQL-based data cleaning, customer-level feature engineering, exploratory data analysis, and machine learning to transform raw retail transactions into actionable customer intelligence.

The main objectives were to:

- Clean and transform raw retail transaction data using SQL
- Apply a time-based historical and future data split
- Engineer customer behavioural features
- Define high-value customers using future spending
- Address class imbalance using SMOTE
- Compare multiple machine learning algorithms
- Optimize Logistic Regression using GridSearchCV
- Evaluate models using Accuracy, Precision, Recall, and F1 Score
- Identify a suitable model for customer-targeting decisions

1.2 Dataset Overview

Dataset: Online Retail Transactions Dataset

The original dataset contained:

- 541,909 transactions
- 8 transaction-level features
- 5,268 duplicate records
- 135,080 missing CustomerID values
- 1,454 missing Description values

After cleaning, temporal splitting, target construction, and customer-level aggregation, the final modelling dataset contained 3,616 customers.

The historical period consisted of transactions before October 1, 2011, while transactions from October 1, 2011 onward were used to calculate future customer spending and create the target variable.

1.3 Target Variable

Future customer spending was used to classify customers into low-value and high-value groups. The median future spending was: £439.61

Customers with future spending above £439.61 were classified as Class 1 — High-Value Customer.

Customers with spending at or below £439.61, including customers with no future purchases, were classified as Class 0 — Low-Value Customer.

The final modelling dataset contained approximately:

- Class 0: 71.65%
- Class 1: 28.35%

This created a moderate class imbalance.

2. DATA CLEANING AND PREPROCESSING

2.1 Data Cleaning

The raw transaction data was loaded into SQLite and preserved in the original table. A separate transactions_clean table was created for analysis.

The cleaning process:

- Removed transactions without CustomerID
- Removed cancelled invoices
- Removed transactions with Quantity ≤ 0
- Removed transactions with UnitPrice ≤ 0
- Removed duplicate records

This approach preserved the original dataset while creating a reproducible analytical dataset.

2.2 Date Standardization

The InvoiceDate field was converted from text into a standardized datetime format: YYYY-MM-DD HH:MM:SS

This enabled reliable chronological processing and the use of SQLite date functions.

2.3 Time-Based Data Partitioning

To reduce the risk of data leakage, the dataset was divided chronologically.

- Historical Data: Transactions before October 1, 2011
- Future Data: Transactions from October 1, 2011 onward

Historical transactions were used to create customer features, while future transactions were used only to calculate future spending and construct the target. This ensured that the model's predictive features were based on information available before the future outcome.

3. FEATURE ENGINEERING AND EXPLORATORY DATA ANALYSIS

3.1 Customer-Level Features

Historical transactions were aggregated at customer level to create seven behavioural features:

- Recency — number of days since the customer's most recent historical purchase
- Frequency — number of distinct invoices associated with the customer
- Average Order Value — average monetary value of customer purchases
- Average Quantity — average quantity purchased
- Total Items — total quantity of products purchased
- Unique Products — number of distinct products purchased
- Number of Transactions — total transaction records associated with the customer

These features captured different dimensions of customer engagement and purchasing behaviour.

3.2 Exploratory Data Analysis

The analysis showed that purchasing-related variables were generally right-skewed, with a relatively small number of customers generating substantially higher purchasing activity.

Outlier analysis identified extreme purchasing values. These observations were retained because they could represent genuine high-value customers rather than data errors, though future iterations should consider log-transforming or winsorizing monetary features to reduce their influence on linear models.

Correlation analysis showed that purchasing-engagement variables such as Frequency, Unique Products, Number of Transactions, and Total Items had positive relationships with the target, while Recency showed a negative relationship.

High-value customers generally demonstrated:

- More frequent purchases
- Greater product diversity
- Higher transaction activity
- Higher purchasing quantities

The customer base was also heavily concentrated in the United Kingdom, representing approximately 90.29% of customers.

Key Insight: Strong historical customer engagement was associated with a greater likelihood of becoming a future high-value customer.

4. BASELINE MACHINE LEARNING MODEL

4.1 Modelling Framework

The baseline model used:

- Algorithm: Logistic Regression
- Training/Test Split: 80% / 20%
- Feature Scaling: StandardScaler
- Random State: 42
- Evaluation Metrics: Accuracy, Precision, Recall, F1 Score

Stratified splitting was used to maintain the approximate class distribution between the training and testing datasets.

4.2 Baseline Performance

The baseline Logistic Regression model achieved:

- Accuracy: 78.73%
- Precision: 72.97%
- Recall: 39.51%
- F1 Score: 51.27%

The model achieved relatively strong Accuracy and Precision but had a low Recall. A Recall of 39.51% means that the model identified only about 40% of customers who actually became high-value customers.

From a business perspective, these missed high-value customers represent potential opportunities lost from targeted marketing, retention, cross-selling, and personalized promotions. Therefore, further optimization was required.

5. CLASS BALANCING AND MODEL OPTIMIZATION

5.1 SMOTE

SMOTE was applied to the training data to address the class imbalance.

Before SMOTE:

- Class 0: 2,072 customers
- Class 1: 820 customers

After SMOTE:

- Class 0: 2,072 customers
- Class 1: 2,072 customers

SMOTE was applied only to the training dataset, while the test dataset remained unchanged.

5.2 Logistic Regression with SMOTE

The SMOTE-based Logistic Regression model achieved:

- Accuracy: 76.52%
- Precision: 57.92%
- Recall: 62.44%
- F1 Score: 60.09%

Compared with the baseline:

- Recall: 39.51% → 62.44%
- F1 Score: 51.27% → 60.09%

Although Accuracy and Precision decreased, the model identified substantially more future high-value customers.

5.3 GridSearchCV Optimization

GridSearchCV was used to optimize the Logistic Regression hyperparameters using five-fold cross-validation and F1 Score as the optimization metric.

The best parameters were:

- C: 0.01
- Penalty: L1
- Solver: liblinear

Best Cross-Validation F1 Score: 72.11%

On the test dataset, the optimized model achieved:

- Accuracy: 74.72%
- Precision: 54.58%
- Recall: 63.90%
- F1 Score: 58.88%

The GridSearchCV model produced the highest Recall among the Logistic Regression configurations, while the standard SMOTE Logistic Regression produced the highest F1 Score.

6. MODEL COMPARISON AND BUSINESS INTERPRETATION

Additional models were evaluated to determine whether alternative algorithms could improve customer-value prediction.

6.1 Model Performance

MODELS	ACCURACY	PRECISION	RECALL	F1
Logistic Regression	78.73%	72.97%	39.51%	51.27%
Logistic Regression + SMOTE	76.52%	57.92%	62.44%	60.09%
Logistic Regression + GridSearchCV	74.72%	54.58%	63.90%	58.88%
Decision Tree	69.06%	45.45%	46.34%	45.89%
Decision Tree + SMOTE	68.09%	44.35%	49.76%	46.90%
Random Forest	78.73%	67.59%	47.80%	56.00%
Random Forest + SMOTE	76.11%	57.69%	58.54%	58.11%
XGBoost	77.35%	62.42%	50.24%	55.68%
XGBoost + SMOTE	75.55%	57.22%	54.15%	55.64%

6.2 Model Interpretation

The results demonstrate a clear trade-off between overall Accuracy and the ability to identify future high-value customers.

The baseline Logistic Regression achieved the highest Precision, while Logistic Regression with SMOTE achieved the highest F1 Score. The GridSearchCV model achieved the highest Recall at 63.90%, meaning it identified the largest proportion of actual future high-value customers among the Logistic Regression approaches. However, its F1 Score of 58.88% was slightly lower than the 60.09% achieved by Logistic Regression with SMOTE.

Therefore, the recommended model depends on the business objective:

- For a balanced combination of Precision and Recall, Logistic Regression + SMOTE is the preferred model.
- If the priority is maximizing the number of potential high-value customers identified, the GridSearchCV model may be preferred because of its higher Recall.

7. CONCLUSION AND RECOMMENDATIONS

7.1 Project Conclusion

This project developed an end-to-end Customer Value Prediction pipeline covering:

- SQL-based data cleaning
- Time-based data partitioning
- Customer-level feature engineering
- Exploratory data analysis
- Machine learning classification
- Class balancing using SMOTE
- Hyperparameter optimization using GridSearchCV
- Comparative model evaluation

The project transformed 541,909 raw transactions into a customer-level modelling dataset containing 3,616 customers and seven behavioural features.

The baseline Logistic Regression model achieved an F1 Score of 51.27%, while Logistic Regression with SMOTE improved the F1 Score to 60.09%. The key improvement was Recall: 39.51% → 62.44%. GridSearchCV further increased Recall to 63.90%.

The results demonstrate that Accuracy alone is not sufficient for evaluating a customer-value prediction problem. Recall and F1 Score are particularly important when missing potential high-value customers represents a significant business opportunity cost.

7.2 Business Applications

The model can support:

- Targeted Marketing: Prioritize customers with higher predicted future value
- Customer Retention: Identify customers for loyalty and retention campaigns
- Cross-Selling: Identify customers with broader purchasing behaviour
- Personalized Promotions: Develop customer segments based on predicted value
- Marketing Resource Allocation: Focus limited resources on high-potential customers

The model should be used as a decision-support and customer-prioritization tool, rather than as the sole basis for marketing decisions.

7.3 Future Improvements

Future work could include:

- Testing additional algorithms such as LightGBM and CatBoost

- Optimizing classification thresholds
- Evaluating ROC-AUC and Precision-Recall AUC
- Engineering additional features such as spending trends and customer lifetime value
- Applying SHAP or other explainability techniques
- Reducing the influence of extreme monetary outliers through log-transformation or winsorization
- Validating the model on another time period or retail dataset

Final Recommended Model: Logistic Regression + SMOTE

Accuracy: 76.52% | Precision: 57.92% | Recall: 62.44% | F1 Score: 60.09%

Final Outcome: The project provides a practical foundation for a customer-targeting decision-support system capable of helping businesses identify potential high-value customers and improve marketing prioritization.